CHAPTER 7

LARGE SAMPLE TESTS

CHAPTER ESSENCE AT A GLANCE:
Master asymptotic Z-tests for large samples (n >= 30): test for single mean, difference of two means, single proportion, and difference of two proportions.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Large Sample Rationale (n >= 30):** Based on the Central Limit Theorem, the sampling distributions of sample statistics are approximately normal; unknown population parameters can be replaced by sample estimates.

• **Test for Single Mean and Difference:** Z = (x_{bar} - mu0) divided by (sigma divided by sqrt(n)); Two-sample Z = (x_{bar1} - x_{bar2}) divided by sqrt($s1^2$ divided by n1 + $s2^2$ divided by n2).

• **Test for Single and Two Proportions:** Single proportion Z = (p - P0) divided by sqrt(P0*Q0 divided by n); Two proportions use pooled estimate P_{hat} = (n1*p1 + n2*p2) divided by (n1 + n2).

• **Critical Decision Boundaries:** Two-tailed test at 5 percent rejects H0 if |Z| > 1.96; at 1 percent rejects H0 if |Z| > 2.58; one-tailed test at 5 percent uses Z = 1.645.

ESSENTIAL STATISTICAL FORMULAS & MEASURES

$$Z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}} \sim N(0,1) \quad [\text{Test for Single Mean}]$$

Large Sample Single Mean Z-Test Statistic

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \sim N(0,1) \quad [\text{Difference of Means}]$$

Two-Sample Independent Means Comparison Z-Statistic

$$Z = \frac{p - P_0}{\sqrt{\frac{P_0 Q_0}{n}}} \sim N(0,1) \quad [\text{Test for Single Proportion}]$$

Large Sample Single Proportion Z-Test Statistic

$$\hat{P} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} \rightarrow Z = \frac{p_1 - p_2}{\sqrt{\hat{P}\hat{Q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

Pooled Two-Sample Proportions Difference Test Statistic

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** In testing a single proportion, use P0 and Q0 (from null hypothesis H0) in the denominator standard error, NOT the sample proportion p.

• **Alert:** In testing equality of two proportions (H0: P1 = P2), always compute the pooled proportion P_{hat} for the standard error.

• **Alert:** Remember that large sample theory applies when n >= 30; if n < 30, use Student's t-test instead.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. A sample of 100 tyres has an average life of 15,200 km with a standard deviation of 800 km. Can it be regarded as a random sample from a population with an average life of 15,000 km? Test at 5 percent level of significance. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Hypotheses:** Null Hypothesis H0: mu = 15,000 km. Alternative Hypothesis H1: mu != 15,000 km (Two-tailed test). **[1 Mark]**

• **Parameters:** n = 100, x_{bar} = 15,200, s = 800. Level of significance α = 0.05. Critical value Z = +- 1.96. **[1 Mark]**

• **Calculated Z:** Z = (x_{bar} - μ) divided by (s divided by square root of n) = (15,200 - 15,000) divided by (800 divided by 10) = 200 divided by 80 = 2.50. **[1 Mark]**

• **Conclusion:** Since calculated |Z| = 2.50 > 1.96, it falls in the rejection region. We reject H0 at 5 percent level. The sample cannot be regarded as drawn from a population with mean life 15,000 km. **[1 Mark]**

2. An examination was given to two groups of students from two colleges: College A: n1 = 50, Mean1 = 75, s1 = 8; College B: n2 = 60, Mean2 = 72, s2 = 6. Test whether there is a significant difference between the mean marks of the two colleges at 5 percent level of significance. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Hypotheses:** H0: mu1 = mu2 (no significant difference in mean marks). H1: mu1 != mu2 (two-tailed test). alpha = 0.05, critical Z = +- 1.96. **[1 Mark]**

• **Standard Error:** SE = square root of [$s1^2$ divided by n1 + $s2^2$ divided by n2] = square root of [64 divided by 50 + 36 divided by 60] = square root of [1.28 + 0.60] = square root of 1.88 = 1.371. **[1 Mark]**

• **Calculated Z:** Z = (Mean1 - Mean2) divided by SE = (75 - 72) divided by 1.371 = 3 divided by 1.371 = 2.188. **[1 Mark]**

• **Conclusion:** Since calculated |Z| = 2.19 > 1.96, we reject H0 at 5 percent level. There is a statistically significant difference between the average performance of the two colleges. **[1 Mark]**

3. A coin is tossed 400 times and turns up heads 216 times. Test at 5 percent level of significance whether the coin is unbiased. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Hypotheses:** H0: P = 0.50 (coin is unbiased). H1: P != 0.50 (two-tailed test). alpha = 0.05, critical Z = +- 1.96. **[1 Mark]**

• **Sample Proportion:** p = 216 divided by 400 = 0.54. P0 = 0.50, Q0 = 0.50, n = 400. **[1 Mark]**

• **Calculated Z:** SE = square root of [(P0 * Q0) divided by n] = square root of [(0.5 * 0.5) divided by 400] = square root of [0.25 divided by 400] = 0.5 divided by 20 = 0.025. Z = (0.54 - 0.50) divided by 0.025 = 0.04 divided by 0.025 = 1.60. **[1 Mark]**

• **Conclusion:** Since calculated |Z| = 1.60 < 1.96, it falls in the acceptance region. We fail to reject H0. The coin can be considered unbiased at 5 percent significance level. **[1 Mark]**

4. Write down the formula for testing the significance of difference between two sample proportions and explain how the pooled proportion is calculated. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Test Statistic Formula:** Z = (p1 - p2) divided by square root of [P_{hat} * Q_{hat} * (1 divided by n1 + 1 divided by n2)], where Z follows standard normal N(0, 1) under H0. **[1.5 Marks]**

• **Pooled Proportion Calculation:** P_{hat} = (n1 * p1 + n2 * p2) divided by (n1 + n2) = (x1 + x2) divided by (n1 + n2), and Q_{hat} = 1 - P_{hat}. **[1.5 Marks]**

5. State the decision rule for a one-tailed right-sided Z-test at: (a) 5 percent level, (b) 1 percent level. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **(a) At 5 percent level:** Reject H0 if calculated Z > +1.645. **[1 Mark]**

• **(b) At 1 percent level:** Reject H0 if calculated Z > +2.33. **[1 Mark]**

6. State two basic assumptions underlying large sample Z-tests. **[Score: 2 Marks]**

Answer & Key Valuation Steps:

• **Assumption 1:** The sample is drawn randomly and units are statistically independent. **[1 Mark]**

• **Assumption 2:** Sample size is sufficiently large (n >= 30) so that the Central Limit Theorem ensures normality of the test statistic. **[1 Mark]**